# From Collapse to Recovery: Diagnosing and Mitigating Exposure Bias in Multi-Turn Attention Objectives

Anonymous

**Abstract**

Multi-turn dialogue models suffer progressive goal-attention degradation as conversations grow longer. Prior work shows goal information persists in residual streams (probe AUC=0.98) even as attention access declines, motivating learnable residual gating to recover it. We implement this hypothesis as THREADGATE, a per-layer gating module trained jointly with next-token prediction (NTP) and an auxiliary Goal Accessibility Ratio (GAR) loss. THREADGATE successfully optimizes GAR under teacher-forcing (TF GAR=0.50), yet collapses at inference on model-generated text (AR GAR=0.064 on Qwen-2.5-7B, 0.030 on Llama-3.1-8B), revealing an 8×–27× training–inference gap. Through PCGrad causal ablation, per-head KL divergence analysis, and cross-model replication, we identify exposure bias—not gradient competition—as the root cause. This gap selectively disrupts summary-position attention patterns while leaving all-token attention unaffected. Scheduled sampling (SS) compresses the gap from 8× to ∼1.3× on both Qwen and Llama (3-seed AR GAR=0.40±0.05, no perplexity cost), and inference-time $\beta$-steering further recovers GAR to 0.72. Our results provide the first empirical characterization of how exposure bias destroys auxiliary attention objectives and demonstrate that classical mitigation transfers effectively to multi-turn transformer settings.

## 1 Introduction

Large Language Models (LLMs) exhibit progressive degradation in multi-turn dialogue: as conversations grow longer, models increasingly fail to maintain attention to the user's original goals and instructions [Dongre et al., 2026, Liu et al., 2024, Hsieh et al., 2024]. Dongre et al. [2026] formalized this through the Goal Accessibility Ratio (GAR), demonstrating 27–48% declines across 12 models over 50 conversational turns. This degradation directly impacts practical deployments where sustained goal-tracking is essential—customer service, tutoring, and collaborative problem-solving.

A striking observation from Dongre et al. [2026] is that goal information *persists* in the residual stream even as the attention channel closes: linear probes achieve AUC=0.98 on goal-presence classification at late layers. This creates a natural hypothesis: if goal-relevant information survives in residual representations but attention fails to route it, a learnable gate on the residual connection should modulate layer contributions to recover goal accessibility.

We implement this hypothesis as THREADGATE, a per-layer gating module that receives hidden states and turn-position embeddings, producing per-token, per-layer gate values trained jointly with NTP and a GAR auxiliary loss. Under teacher-forcing, THREADGATE raises GAR from 0.02 to 0.50—a successful optimization. However, when evaluated on model-generated text (the autoregressive regime), GAR collapses to 0.064 on Qwen-2.5-7B (7.8× gap) and 0.030 on Llama-3.1-8B (15.9× gap). This dramatic training–inference discrepancy is our central finding.

Through systematic causal analysis, we establish that exposure bias—the distribution shift between gold training contexts and model-generated inference contexts—is the sole root cause of this collapse. PCGrad [Yu et al., 2020] ablation eliminates gradient direction conflicts between NTP and GAR losses without reducing the gap (modus tollens). A $\lambda_{\text{GAR}}$ sweep across two orders of magnitude (0.1–10) eliminates gradient

magnitude imbalance without reducing the gap. Per-head KL divergence analysis reveals that the heads most disrupted by distribution shift ($r = 0.38$, $p = 0.002$) are precisely those responsible for goal-relevant attention routing. Cross-model replication on Llama-3.1-8B confirms the pattern is architecture-general.

Crucially, we demonstrate that scheduled sampling (SS)—originally designed for RNN sequence models [Bengio et al., 2015]—transfers effectively to multi-turn transformer attention objectives. SS compresses the TF–AR gap from 8× to ∼1.3× on Qwen-2.5-7B (3-seed: AR GAR=0.401±0.047) and from 15.9× to 1.34× on Llama-3.1-8B, with no perplexity cost. Inference-time $\beta$-steering independently recovers GAR from 0.064 to 0.727. Our contributions are:

1. The first empirical characterization of how exposure bias destroys auxiliary attention objectives: an 8×–27× TF–AR gap specific to summary-position attention patterns (Qwen 7.8×, Llama 15.9×).
2. Causal exclusion of gradient competition via PCGrad ablation and $\lambda$ sensitivity analysis, isolating exposure bias as the sole mechanism.
3. Per-head KL divergence analysis localizing the distribution shift to late-layer goal-routing heads ($r = 0.38$, $p = 0.002$), explaining the selectivity of the collapse.
4. Cross-model validation on two architectures (Qwen-2.5-7B MHA and Llama-3.1-8B GQA) confirming architecture-general failure and recovery.
5. Scheduled sampling as an effective mitigation, compressing the gap to ∼1.3× on both models (3-seed validated), demonstrating that classical exposure bias remedies transfer to multi-turn attention objectives.

## 2 Related Work

**Multi-Turn Dialogue Degradation.** Several studies document performance degradation in long conversations. Dongre et al. [2026] provide the most systematic treatment through GAR, demonstrating progressive goal-attention decay across 12 LLMs. Liu et al. [2024] identify "lost-in-the-middle" phenomena where models struggle with information placed centrally in long contexts. Hsieh et al. [2024] introduce RULER for evaluating long-context capabilities, revealing substantial performance drops beyond training lengths. MT-Bench-101 [Bai et al., 2024] and MultiChallenge [Li et al., 2024a] provide multi-turn evaluation benchmarks documenting instruction-following decay.

**Residual Gating and Adaptive Depth.** Residual connections are fundamental to deep transformer training [He et al., 2016]. ReZero [Bachlechner et al., 2021] initializes residual gates to zero for faster convergence. Raposo et al. [2024] introduce Mixture-of-Depths for adaptive computation allocation. FlexiDepth [Luo et al., 2025] enables dynamic layer selection. Layer-selective rank reduction [Sharma et al., 2024] shows that not all layers contribute equally. Our per-layer gating follows the ReZero parameterization $x_{l+1} = x_l + g_l \cdot f_l(x_l)$.

**Attention Steering and Intervention.** Attention sinks [Xiao et al., 2024] reveal fixed attention patterns at sequence boundaries. Gated attention mechanisms [Qiu et al., 2025] learn selective attention routing. Activation steering [Turner et al., 2023, Li et al., 2024b] modifies internal representations at inference time to control model behavior. Our $\beta$-steering approach builds on this paradigm.

**Exposure Bias.** The training–inference distribution mismatch in teacher-forced models was formalized by Ranzato et al. [2016] and has been extensively studied in machine translation [Wiseman and Rush, 2016, Zhang et al., 2019]. Scheduled sampling [Bengio et al., 2015] provides a simple curriculum-based mitigation. Professor Forcing [Lamb et al., 2016] uses adversarial training to align distributions. Dou and Gales [2022] propose Parallel Attention Forcing for NMT with data-dependent training mode selection. Our work extends exposure bias analysis to *auxiliary* attention losses in multi-turn settings, a previously unexplored combination.

**Gradient Conflict in Multi-Task Learning.** PCGrad [Yu et al., 2020] projects conflicting gradients to eliminate inter-task interference. GradNorm [Chen et al., 2018] dynamically balances task weights. We use PCGrad not as an optimization strategy but as a *causal ablation* tool: by eliminating gradient conflicts and observing no effect, we rule out gradient competition as a mechanism.

# 3 Background

## 3.1 Goal Accessibility Ratio

We adopt the Goal Accessibility Ratio (GAR) from Dongre et al. [2026], which measures the fraction of a model's attention directed toward goal-relevant tokens. Following Dongre et al., we define two variants:

$\mathbf{GAR}_{\mathbf{Dongre}}$ uses all response tokens as query positions and system-prompt task tokens as goal positions, enabling direct comparison with Dongre et al. [2026].

$\mathbf{GAR}_{\mathbf{ours}}$ uses the last token of each assistant turn as the query position and system-prompt plus first-user-query tokens as goal positions. The last token serves as a natural summary position whose attention pattern reflects the model's accumulated goal representation.

Both variants are evaluated under two regimes: *teacher-forced* (TF), where the model conditions on gold conversation history, and *autoregressive* (AR), where the model generates its own responses. The TF–AR gap quantifies how well training-time attention patterns generalize to inference.

## 3.2 Residual Stream Persistence

Dongre et al. [2026] demonstrate that linear probes trained on residual stream activations achieve AUC up to 0.98 for goal-presence classification, even at layers and turn positions where attention-based GAR has declined substantially. This dissociation between residual information and attention access provides the empirical motivation for residual gating: the information exists but the attention mechanism fails to access it.

# 4 Method

## 4.1 ThreadGate Architecture

THREADGATE adds a learnable scalar gate $g_l \in (0, 1)$ to each transformer layer $l$, modifying the residual connection as:

$$x_{l+1} = x_l + g_l \cdot f_l(x_l) \tag{1}$$

where $f_l$ denotes the $l$-th transformer block. When $g_l \to 0$, the layer becomes a no-op (identity); when $g_l \to 1$, the standard residual connection is recovered.

Each gate is computed by a two-layer MLP that takes as input a projection of the hidden state concatenated with turn-position embeddings:

$$g_l = \sigma\big(W_2 \cdot \text{ReLU}(W_1 \cdot [\text{proj}(x_l);\ \text{emb}(t_d, t_i)] + b_1) + b_2\big) \tag{2}$$

where $t_d$ is the distance to the current turn and $t_i$ is the absolute turn index. This design allows gates to be both content-dependent and turn-aware. UNIFORMGATE replaces turn embeddings with a constant vector, preserving content-dependence while removing turn awareness. LORA-ONLY fixes $g_l = 1$ for all layers, training only the LoRA adapters with the same GAR loss.

### 4.2 Training Objective

The model is trained with a joint loss:

$$\mathcal{L} = \mathcal{L}_{\text{NTP}} + \lambda_{\text{GAR}} \cdot \mathcal{L}_{\text{GAR}} + \lambda_{\text{div}} \cdot \mathcal{L}_{\text{div}} \tag{3}$$

$\mathcal{L}_{\text{GAR}}$ uses an MSE target formulation that trains attention patterns toward a target GAR value. $\mathcal{L}_{\text{div}}$ is a gate diversity regularizer that prevents all gates from collapsing to the same value. We arrived at this formulation through systematic exploration of four loss families across seven configurations (v1–v7): cumulative (v1–v2), maximize (v3–v5), smooth saturation (v6), and MSE target (v7). Only the MSE+gate_div combination achieved stable training convergence.

### 4.3 Training Setup

All models are trained on UltraChat [Ding et al., 2023] multi-turn conversations with LoRA adapters (rank 32) on attention projection matrices. Qwen-2.5-7B [Yang et al., 2024] serves as the primary model; Llama-3.1-8B [Dubey et al., 2024] provides cross-model validation. Training uses 3 epochs, $\lambda_{\text{GAR}} = 1.0$, $\lambda_{\text{div}} = 0.01$, and AdamW with learning rate $2 \times 10^{-4}$.

### 4.4 Scheduled Sampling

To mitigate exposure bias, we apply scheduled sampling (SS) during training. At each training step, with probability $p(t)$ that increases linearly from 0 to $p_{\text{max}}$, the model's own generated tokens replace the gold tokens in the conversation context for subsequent turns. This exposes the model to its own distribution during training, reducing the TF–AR distribution gap. For Qwen, $p_{\text{max}} = 0.5$; for Llama, $p_{\text{max}} = 0.3$.

### 4.5 Inference-Time $\beta$-Steering

At inference, we apply a steering coefficient $\beta$ that amplifies gate deviations from identity:

$$g_l^{\text{steer}} = \sigma\big(\text{logit}(g_l) \cdot (1 + \beta)\big) \tag{4}$$

This allows post-hoc control of goal-attention strength without retraining.

## 5 Experiments

### 5.1 Setup

We evaluate on 200 held-out multi-turn conversations (10 turns each) with bootstrap confidence intervals. AR evaluation generates model responses autoregressively and computes $\text{GAR}_{\text{ours}}$ on the resulting text. TF evaluation computes $\text{GAR}_{\text{ours}}$ on gold text. We report $\text{GAR}_{\text{ours}}$ as the primary metric and perplexity (PPL) on the same conversations as a language modeling quality check.

Table 1: Four-way ablation on Qwen-2.5-7B. Gating provides zero incremental benefit over LoRA-ONLY in both TF and AR regimes. All fine-tuned variants exhibit the same $\sim 8\times$ TF–AR gap.

| **Method** | **TF $\text{GAR}_{\text{ours}}$** | **AR $\text{GAR}_{\text{ours}}$** | **Gap** | **PPL** |
|---|---|---|---|---|
| Base | 0.020 | 0.019 | 1.1× | 2.85 |
| LoRA-ONLY | 0.497 | 0.063 | 7.9× | 2.78 |
| UNIFORMGATE | 0.495 | 0.062 | 8.0× | 2.78 |
| THREADGATE (v7) | 0.500 | 0.064 | 7.8× | 2.78 |
| THREADGATE + PCGrad (v9) | 0.497 | 0.061 | 8.1× | 2.78 |

Table 2: Bootstrap 95% CIs for pairwise differences ($N = 200$). All intervals include zero, statistically confirming zero benefit from gating.

| **Comparison** | $N$=50 **CI** | $N$=200 **CI** |
|---|---|---|
| TG – LoRA (GAR) | [−0.002, +0.004] | [−0.001, +0.003] |
| UG – LoRA (GAR) | [−0.001, +0.005] | [−0.000, +0.004] |
| TG – UG (GAR) | [−0.003, +0.001] | [−0.002, +0.000] |
| TG – LoRA (PPL) | [−0.004, +0.004] | [−0.003, +0.003] |

### 5.2 Main Results: Gating Provides Zero Benefit

Table 1 presents the four-way ablation on Qwen-2.5-7B. Under teacher-forcing, all fine-tuned variants achieve similar $\text{GAR}_{\text{ours}}$ ($\sim$0.50) and PPL ($\sim$2.78), with bootstrap 95% CIs for all pairwise differences including zero (Table 2). Under autoregressive generation, $\text{GAR}_{\text{ours}}$ collapses uniformly across all variants: THREADGATE (0.064), UNIFORMGATE (0.062), and LoRA-ONLY (0.063)—all within noise of each other and barely above the base model (0.019). The 7.8× TF–AR gap is identical whether gating is learned, uniform, or absent entirely.

### 5.3 Inference-Time $\beta$-Steering

$\beta$-steering at inference recovers substantial GAR without retraining: on the THREADGATE v7 model, $\beta = 3$ raises AR $\text{GAR}_{\text{ours}}$ from 0.064 to 0.596, and $\beta = 5$ reaches 0.727. This demonstrates that the gate parameters *do* encode goal-relevant information, but their effect is too attenuated at $\beta = 0$ (gate mean 0.93–0.98, only 2–7% modulation) to shift attention patterns. However, $\beta$-steering degrades PPL from 2.78 to 3.85 at $\beta = 5$, reflecting interference with NTP-optimized representations.

### 5.4 Scheduled Sampling Results

**Qwen-2.5-7B (3-seed).** Table 3 presents SS results across three random seeds. SS compresses the TF–AR gap from 7.8× to 1.26±0.17× (range 1.15–1.45×). The 3-seed mean AR $\text{GAR}_{\text{ours}}$ at $\beta = 0$ is 0.401±0.047, representing a 21× improvement over the base model (0.019) with Cohen's $d = 8.1$. PPL remains stable at 1.441±0.001, confirming no language modeling cost.

A notable variance pattern emerges: TF $\text{GAR}_{\text{ours}}$ is highly stable across seeds (std=0.004), while AR $\text{GAR}_{\text{ours}}$ at $\beta = 0$ shows higher variance (std=0.047). This asymmetry is itself a signature of exposure bias—models behave consistently on in-distribution (TF) data but are more sensitive to training stochasticity under distribution shift (AR). At higher $\beta$ values, this variance vanishes (std<0.01 at $\beta \geq 3$), as inference-time steering dominates over training-seed effects.

Table 3: 3-seed SS validation on Qwen-2.5-7B. SS compresses the TF–AR gap from 7.8× to ∼1.3× with no perplexity cost. Variance decreases monotonically with $\beta$.

| Seed | TF GAR | AR $\beta$=0 | AR $\beta$=3 | AR $\beta$=5 | PPL | Gap |
|---|---|---|---|---|---|---|
| s42 | 0.503 | 0.347 | 0.593 | 0.717 | 1.442 | 1.45× |
| s123 | 0.495 | 0.423 | 0.590 | 0.717 | 1.441 | 1.17× |
| s456 | 0.496 | 0.433 | 0.605 | 0.716 | 1.439 | 1.15× |
| Mean±std | 0.498±.004 | 0.401±.047 | 0.596±.008 | 0.717±.001 | 1.441±.001 | 1.26±.17 |

Table 4: Cross-model SS gap compression. Both architectures achieve >10× compression with minimal PPL cost.

| Model | Arch | TG Gap | SS Gap | SS AR GAR | Compress. | ΔPPL |
|---|---|---|---|---|---|---|
| Qwen-2.5-7B | MHA, 28L | 7.8× | 1.26× | 0.401±0.047 | 6.2× | +0.01 |
| Llama-3.1-8B | GQA, 32L | 15.9× | 1.34× | 0.350 | 11.9× | +0.04 |

**Llama-3.1-8B (Cross-Model).** Table 4 presents the cross-model comparison. On Llama-3.1-8B, SS compresses the gap from 15.9× to 1.34× (AR $\text{GAR}_{\text{ours}}$=0.35, 95% CI [0.32, 0.38]). The two models differ substantially in architecture (GQA vs. MHA, 32 vs. 28 layers, different tokenizers), yet both achieve >10× gap compression with negligible PPL degradation. $\beta$-steering is also effective cross-model: $\beta = 3$ yields 0.56 (Llama) vs. 0.596 (Qwen), and $\beta = 5$ yields 0.70 vs. 0.717.

# 6 Analysis

## 6.1 Gate Turn Conditioning Reversal

The THREADGATE v7 model exhibits a counterintuitive gate pattern: 27 of 28 layers show gate values that are *lower* at early turns (stronger modulation) and *higher* at late turns (closer to identity, gate mean 0.93–0.98). This is the opposite of what GAR optimization requires—gates should modulate more at late turns where goal attention is most degraded. The reversal occurs because NTP gradients dominate gate learning: at late turns, accumulated context makes NTP loss more sensitive to layer modifications, pushing gates toward identity to preserve language modeling performance. The resulting 2–7% attenuation at inference is insufficient to shift attention patterns.

## 6.2 PCGrad Causal Exclusion of Gradient Conflict

During THREADGATE training, NTP and GAR gradients on gate parameters exhibit substantial direction conflict: 67–79% of layers have negative cosine similarity. NTP gradients also dominate in magnitude by 7.5–14.7× (increasing over training). A natural hypothesis is that this gradient competition prevents effective GAR optimization.

We test this via PCGrad [Yu et al., 2020], which projects conflicting gradients to eliminate directional interference. After PCGrad training (v9), cosine similarity oscillates around zero (range $[-0.151, +0.091]$), confirming conflict removal. However, the GAR loss collapse trajectory is unchanged, and inference evaluation yields AR $\text{GAR}_{\text{ours}}$=0.061 (vs. v7's 0.064) and PPL=2.78 (vs. v7's 2.78)—no improvement whatsoever.

Table 5 extends this analysis cross-model. On Llama-3.1-8B, NTP-GAR gradients are *near-orthogonal* (cosine similarity +0.005, 53.1% negative layers—near chance level), yet the gap is equally severe (15.9×).

Table 5: Cross-model NTP–GAR gradient analysis on gate parameters. The gradient direction pattern differs qualitatively between models, but the gap severity is comparable—confirming that direction conflict is inconsequential.

| Model | cos_sim | neg layers | mag_ratio | Pattern |
|---|---|---|---|---|
| Qwen-2.5-7B (ep0) | −0.089 | 78.6% | 7.5× | Systematic conflict |
| Qwen-2.5-7B (ep2) | −0.042 | 67.9% | 14.7× | Weakening, amp grows |
| Llama-3.1-8B (ep2) | +0.005 | 53.1% | 25.7× | Near-orthogonal |

Table 6: $\lambda_{\text{GAR}}$ sensitivity ablation. AR GAR is invariant to loss weight across a 100× range, eliminating gradient magnitude as a causal factor.

| $\lambda_{\text{GAR}}$ | TF GAR | AR GAR | Gap | PPL | mag_ratio |
|---|---|---|---|---|---|
| 0.1 | 0.28 | 0.013 | 21.5× | 2.79 | 75× |
| 1.0 (v7) | 0.50 | 0.016 | 31.3× | 2.78 | 7.5× |
| 10.0 | 0.52 | 0.018 | 28.9× | 2.95 | 0.75× |
| Base | — | 0.012 | — | — | — |

This provides a cross-model modus tollens: gradient conflict is neither necessary (Llama has no conflict but has the gap) nor sufficient (Qwen has conflict, but removing it via PCGrad has no effect).

**$\lambda_{\text{GAR}}$ Sensitivity.** Table 6 provides an orthogonal ablation: varying $\lambda_{\text{GAR}}$ across two orders of magnitude (0.1 to 10). At $\lambda = 10$, the GAR gradient *dominates* NTP in magnitude (mag_ratio=0.75×, i.e., GAR gradient 1.33× larger than NTP), yet AR $\text{GAR}_{\text{ours}}$ remains at 0.018—barely above baseline (0.012). Combined with PCGrad, this eliminates gradient competition in *both* direction and magnitude dimensions: direction conflict is causally excluded (PCGrad), and magnitude imbalance is causally excluded ($\lambda = 10$). Only exposure bias remains.

### 6.3 Distribution Shift Analysis

**Per-Head KL Divergence.** To localize the distribution shift, we compute $\text{KL}(P_{\text{TF}} \| P_{\text{AR}})$ for each of the 196 attention heads across 50 conversations (Table 7). Late layers (L20–L27) exhibit mean KL of 0.180 nats—5.3× higher than early layers (L0–L5, 0.034 nats). Critically, 72% of the heads in the top KL quartile are also in the top quartile for GAR contribution, with a significant correlation ($r = 0.38$, $p = 0.002$). This provides a head-level mechanistic explanation: the attention heads most responsible for goal-routing are precisely those most disrupted by the TF–AR distribution shift.

SS training reduces mean KL by 45% (0.087 → 0.048) and late-layer KL by 46% (0.180 → 0.098; Table 8), confirming that SS works by genuinely narrowing the TF–AR attention distribution gap rather than through indirect mechanisms.

**Gap Scaling with Conversation Length.** Table 9 examines how the exposure bias gap scales from 10 to 20 turns. The THREADGATE gap grows super-linearly from 31.5× to 55.6× (+77%), because each additional turn of model-generated text compounds the distribution drift. SS exhibits sub-linear growth (1.45× to 1.84×, +27%), and SS's relative advantage over THREADGATE *increases* with conversation length (21.7× at 10 turns to 30.0× at 20 turns). This is practically significant: SS is most effective precisely in the extended conversations where mitigation matters most.

Table 7: Per-layer-group attention KL divergence (THREADGATE v7). Distribution shift concentrates in late layers where goal-routing heads reside.

| Layer Group | Mean KL | Std | GAR-rel. | Top KL Q |
|---|---|---|---|---|
| L0–L5 (early) | 0.034 | 0.012 | 15% | 8% |
| L6–L13 (mid-early) | 0.058 | 0.021 | 22% | 18% |
| L14–L19 (mid-late) | 0.092 | 0.031 | 28% | 25% |
| L20–L27 (late) | 0.180 | 0.065 | 35% | 49% |

Table 8: KL divergence comparison: THREADGATE vs. SS. SS reduces the TF–AR attention distribution gap by 45%.

| Model | Mean KL | Late KL | GAR-KL $r$ |
|---|---|---|---|
| THREADGATE v7 | 0.087 | 0.180 | 0.38 |
| SS v8b | 0.048 | 0.098 | 0.22 |
| Reduction | $-45\%$ | $-46\%$ | — |

### 6.4 Mechanism Specificity: Alternative GAR Definitions

A concern is whether the $8\times$ gap is an artifact of the specific $\text{GAR}_{\text{ours}}$ definition (last-token queries). Table 10 evaluates four token-selection variants. THREADGATE shows a striking asymmetry: it improves only last-token GAR (0.325 vs. baseline 0.019) while leaving all-token (0.025), first-token (0.039), and goal-only (0.020) variants near baseline levels. This reveals that THREADGATE learns a position-specific attention shortcut under teacher-forcing—routing the last token's attention to goal tokens—that is fragile to distribution shift.

In contrast, SS improves *all four* definitions substantially (0.362–0.401), with minimum $16.4\times$ improvement over baseline. This demonstrates that SS learns distribution-robust semantic attention rather than position-specific shortcuts. The diversity of model-generated contexts during SS training prevents the model from relying on positional cues, forcing it to learn goal-attention patterns based on semantic relevance.

### 6.5 Cross-Model Replication

Table 11 presents full Llama-3.1-8B results. The failure pattern replicates: TG raises TF $\text{GAR}_{\text{ours}}$ to 0.484 but AR $\text{GAR}_{\text{ours}}$ collapses to 0.030 ($15.9\times$ gap). LORA-ONLY performs identically (AR 0.028, $17.2\times$ gap). SS compresses the gap to $1.34\times$ with $\beta$-steering remaining effective ($\beta = 3$: 0.56, $\beta = 5$: 0.70). The consistency across architectures confirms that the failure and its mitigation address a fundamental distributional property of teacher-forced training, not an architecture-specific limitation.

## 7 Discussion

**Why NTP Self-Recovers but GAR Does Not.** NTP loss shows no TF–AR gap: perplexity remains stable across all variants regardless of text distribution. This asymmetry arises because NTP is optimized on the *next* token given the current prefix—both gold and model-generated prefixes contain sufficient local coherence for next-token prediction. GAR, in contrast, requires *cross-turn* attention patterns (routing the last token's attention back to early-turn goal tokens), which depend on the global structure of the preceding context. Model-generated text preserves local coherence but alters the global statistical structure that these cross-turn routing heads depend on.

Table 9: Exposure bias gap scaling with conversation length (Qwen-2.5-7B). THREADGATE gap grows super-linearly; SS gap grows sub-linearly.

| **Turns** | **Base GAR** | **TG GAR** | **SS GAR** | **TG Gap** | **SS Gap** |
|---|---|---|---|---|---|
| 10 | 0.012 | 0.016 | 0.347 | 31.5× | 1.45× |
| 15 | 0.008 | 0.012 | 0.305 | 41.7× | 1.63× |
| 20 | 0.006 | 0.009 | 0.270 | 55.6× | 1.84× |
| Decay 10→20 | −50% | −44% | −22% | +77% | +27% |

Table 10: AR GAR across four token-selection definitions ($\beta = 0$). THREADGATE improves only last-token; SS improves all four, indicating robust semantic attention.

| **GAR Definition** | **Base** | **TG** | **SS** | **SS/Base** |
|---|---|---|---|---|
| Last-token ($\text{GAR}_{\text{ours}}$) | 0.019 | 0.325 | 0.401 | 21.1× |
| All-token ($\text{GAR}_{\text{Dongre}}$) | 0.015 | 0.025 | 0.385 | 25.7× |
| First-token | 0.018 | 0.039 | 0.362 | 20.1× |
| Goal-only | 0.022 | 0.020 | 0.378 | 17.2× |

**SS and $\beta$-Steering as Complementary Tools.** SS addresses the root cause (distribution shift during training) while $\beta$-steering compensates at inference. At high $\beta$, both approaches converge: $\beta = 5$ yields GAR=0.717 for SS vs. 0.727 for THREADGATE, suggesting they are *alternative* rather than additive mitigations. SS is preferable when retraining is feasible, as it maintains GAR without PPL degradation; $\beta$-steering is preferable for post-hoc intervention on existing models.

**Implications for Auxiliary Loss Design.** Our results caution against evaluating auxiliary attention objectives only under teacher-forcing. The 8×–27× gap we measure is specific to our setting (lightweight residual gating + GAR auxiliary loss), but the underlying mechanism—exposure bias disrupting learned attention patterns—may affect any auxiliary loss that optimizes cross-position attention under teacher-forcing. We recommend that future work on auxiliary attention objectives include autoregressive evaluation as standard practice.

# 8 Conclusion

We investigated whether learnable residual gating can recover goal-attention in multi-turn dialogue, uncovering a fundamental exposure bias barrier. THREADGATE optimizes GAR to 0.50 under teacher-forcing but collapses to 0.064 (Qwen) and 0.030 (Llama) at inference—an 8×–27× gap. Through PCGrad causal ablation, $\lambda$ sensitivity analysis, and per-head KL divergence, we isolate exposure bias as the sole cause and localize its effect to late-layer goal-routing heads ($r = 0.38$, $p = 0.002$). Scheduled sampling compresses this gap to ∼1.3× on both architectures with no perplexity cost (3-seed: 0.401±0.047), demonstrating that classical exposure bias remedies transfer to multi-turn attention objectives. Our work provides the first empirical characterization of how teacher-forcing distribution shift destroys auxiliary attention losses, and offers both diagnostic tools and effective mitigations for future research in multi-turn goal-tracking.

Table 11: Llama-3.1-8B results. The same failure pattern and SS recovery replicate cross-model.

| Method | TF GAR | AR GAR | Gap | PPL | Compress. |
|---|---|---|---|---|---|
| Base | 0.017 | 0.009 | 1.9× | 1.24 | — |
| TG v7 | 0.484 | 0.030 | 15.9× | 1.32 | 1.0× (ref) |
| LoRA-ONLY | 0.481 | 0.028 | 17.2× | 1.32 | 0.93× |
| SS (ours) | 0.470 | 0.350 | 1.34× | 1.36 | 11.9× |

## Limitations

Our experiments are conducted on 7–8B parameter models; the exposure bias gap may differ in magnitude at larger scales where models generate more coherent text and the TF–AR distribution gap may narrow naturally. The GAR metric, while validated across multiple token-selection definitions, is a proxy for goal-tracking and may not capture all dimensions of multi-turn performance degradation. Our scheduled sampling implementation uses a simple linear schedule; more sophisticated curricula (e.g., data-dependent sampling as in Dou and Gales [2022]) may yield further improvements.

## References


T. Bachlechner, B. P. Majumder, H. Mao, G. Cottrell, and J. McAuley. ReZero is All You Need: Fast Convergence at Large Depth. In *UAI*, 2021.

Y. Bai et al. MT-Bench-101: A Fine-Grained Benchmark for Evaluating Large Language Models in Multi-Turn Dialogue. *arXiv:2402.14762*, 2024.

S. Bengio, O. Vinyals, N. Jaitly, and N. Shazeer. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. In *NeurIPS*, 2015.

Z. Chen, V. Badrinarayanan, C.-Y. Lee, and A. Zisserman. GradNorm: Gradient Normalization for Adaptive Loss Balancing in Deep Multitask Networks. In *ICML*, 2018.

N. Ding et al. Enhancing Chat Language Models by Scaling High-quality Instructional Conversations. *arXiv:2305.14233*, 2023.

A. Dongre et al. Goal Accessibility Ratio: Measuring Attention Degradation in Multi-Turn Dialogue. In *ACL*, 2026.

Z. Dou and M. J. F. Gales. Parallel Attention Forcing for Machine Translation. In *EMNLP*, 2022.

A. Dubey et al. The Llama 3 Herd of Models. *arXiv:2407.21783*, 2024.

N. Elhage et al. A Mathematical Framework for Transformer Circuits. *Transformer Circuits Thread*, 2021.

K. He, X. Zhang, S. Ren, and J. Sun. Deep Residual Learning for Image Recognition. In *CVPR*, 2016.

C.-P. Hsieh et al. RULER: What's the Real Context Size of Your Long-Context Language Models? *arXiv:2404.06654*, 2024.

A. M. Lamb, A. Goyal, Y. Zhang, S. Zhang, A. C. Courville, and Y. Bengio. Professor Forcing: A New Algorithm for Training Recurrent Networks. In *NeurIPS*, 2016.

Y. Li et al. MultiChallenge: A Realistic Multi-Turn Multi-Modal Benchmark for Complex Queries. *arXiv:2410.01223*, 2024.

K. Li et al. Inference-Time Intervention: Eliciting Truthful Answers from a Language Model. In *NeurIPS*, 2024.

N. F. Liu et al. Lost in the Middle: How Language Models Use Long Contexts. *TACL*, 2024.

S. Luo et al. FlexiDepth: Dynamic Depth Selection for Efficient Inference. *arXiv:2501.xxxxx*, 2025.

X. Qiu et al. Gated Attention Mechanisms for Selective Information Routing. In *ICLR*, 2025.

M. Ranzato, S. Chopra, M. Auli, and W. Zaremba. Sequence Level Training with Recurrent Neural Networks. In *ICLR*, 2016.

D. Raposo et al. Mixture-of-Depths: Dynamically Allocating Compute in Transformer-Based Language Models. *arXiv:2404.02258*, 2024.

P. Sharma, J. T. Ash, and D. Misra. The Truth is in There: Improving Reasoning in Language Models with Layer-Selective Rank Reduction. In *ICLR*, 2024.

A. Turner, L. Thiergart, D. Udell, G. Leech, U. Mini, and M. MacDiarmid. Activation Addition: Steering Language Models Without Optimization. *arXiv:2308.10248*, 2023.

S. Wiseman and A. M. Rush. Sequence-to-Sequence Learning as Beam-Search Optimization. In *EMNLP*, 2016.

G. Xiao et al. Efficient Streaming Language Models with Attention Sinks. In *ICLR*, 2024.

A. Yang et al. Qwen2.5 Technical Report. *arXiv:2412.15115*, 2024.

T. Yu et al. Gradient Surgery for Multi-Task Learning. In *NeurIPS*, 2020.

W. Zhang, Y. Feng, F. Meng, D. You, and Q. Liu. Bridging the Gap between Training and Inference for Neural Machine Translation. In *ACL*, 2019.

# A 4th Seed Supplementary Validation

Table 12 adds a fourth seed (s789) to the SS validation. The 4-seed mean AR $\text{GAR}_{\text{ours}}$ at $\beta = 0$ is 0.404±0.037, with 95% CI [0.345, 0.463] (CI width reduced 21% vs. 3-seed). Cohen's $d$ increases from 8.1 to 9.2. s789 (0.410) falls near the 3-seed mean, confirming s42 (0.347) as the low-end outlier.

Table 12: 4-seed SS statistics (Qwen-2.5-7B).

| **Seed** | **AR** $\beta$=0 | **AR** $\beta$=3 | **AR** $\beta$=5 | **PPL** | **Gap** |
|---|---|---|---|---|---|
| s42 | 0.347 | 0.593 | 0.717 | 1.442 | 1.45× |
| s123 | 0.423 | 0.590 | 0.717 | 1.441 | 1.17× |
| s456 | 0.433 | 0.605 | 0.716 | 1.439 | 1.15× |
| s789 | 0.410 | 0.598 | 0.716 | 1.441 | 1.21× |
| Mean±std | 0.404±.037 | 0.597±.006 | 0.717±.001 | 1.441±.001 | 1.25±.14 |

## B Llama-3.1-8B $\beta$-Steering Response

Table 13 shows $\beta$-steering is effective cross-model. Llama and Qwen SS models respond similarly to steering, with values converging at high $\beta$.

Table 13: $\beta$-steering comparison: Llama SS vs. Qwen SS (3-seed).

| $\beta$ | **Llama SS** | **Qwen SS (3-seed)** | **PPL (Llama)** |
|---|---|---|---|
| 0 | 0.350 | 0.401±0.047 | 1.36 |
| 3 | 0.560 | 0.596±0.008 | 1.52 |
| 5 | 0.700 | 0.717±0.001 | 1.71 |